

Graduate Research Statement

Secure and Robust Federated Learning Under Heterogeneous and Adversarial Clients

Yohannes Alelign Biresaw

BSc in Electrical and Computer Engineering | GPA 3.63/4.0

Haramaya University, Ethiopia | yohannes.sch.ca@gmail.com

Research focus

My intended graduate research focuses on the security and robustness of federated learning when clients differ for legitimate reasons and when some clients behave adversarially. The central challenge is that an honest client trained on a minority or shifted data distribution can produce an update that appears unusual, while a malicious client may deliberately imitate normal variation. A useful defense must therefore resist poisoning without excluding legitimate heterogeneous clients.

Research foundation

I developed a completed federated medical-imaging project using the RSNA Pneumonia Detection Challenge dataset. The work included patient-aware dataset construction, DICOM preprocessing, controlled client partitioning, centralized and federated training, multi-seed evaluation, validation-only model selection, and one frozen final test evaluation. I compared FedAvg and FedProx under IID, moderate non-IID, and severe non-IID label distributions.

The main result was not a universal algorithm winner. Increasing heterogeneity reduced predictive performance. FedProx reduced client and global model-update magnitudes by approximately one-half and achieved small gains in balanced accuracy and F1-score, while FedAvg retained a small PR-AUC advantage. The project showed that optimization stability and predictive generalization are related but not equivalent.

Proposed research problem

Building on this foundation, I propose to study robust federated aggregation under both legitimate heterogeneity and malicious participation. The work will define a clear threat model, implement representative poisoning attacks, compare established robust aggregation methods, and measure not only attack resistance but also the cost imposed on honest minority clients.

How can federated aggregation resist malicious-client attacks while preserving legitimate updates generated by non-IID client data?

Planned contribution

- A reproducible cross-silo benchmark that combines patient-aware medical imaging, controlled non-IID data, and malicious-client participation.
- A comparison of FedAvg with coordinate-wise median, trimmed mean, Multi-Krum, geometric-median or RFA aggregation, and trust-based methods where assumptions are satisfied.
- Evaluation of global discrimination, worst-client performance, attack success, benign-client false rejection, calibration, convergence, and computational overhead.
- An analysis of when robust aggregation mistakes legitimate heterogeneity for malicious behaviour.

- A carefully scoped method extension only if baseline experiments reveal a reproducible failure that existing methods do not address.

Why this problem matters

Robust aggregation is often evaluated by overall accuracy after an attack. That view is incomplete when honest clients represent minority classes, acquisition protocols, or institutions. A defense may appear successful globally while systematically suppressing the clients whose updates are most different from the majority. Measuring benign-client rejection and worst-client performance therefore makes the security evaluation more realistic and more relevant to cross-silo applications.

Research approach and values

I intend to use prespecified attacks, aggregation baselines, seeds, and evaluation criteria. Claims will be tied to observable evidence rather than to a single favourable run. Completed results, proposed experiments, and limitations will remain clearly separated. The goal is not to introduce a new method prematurely, but to identify a reproducible failure and then develop the smallest justified extension.

Preparation and development goals

My Electrical and Computer Engineering background and professional experience in IT support and networking give me a practical systems perspective. Through the completed project, I gained experience with Python, TensorFlow/Keras, medical-image preprocessing, federated optimization, experiment tracking, Git/GitHub, and disciplined evaluation. Graduate supervision will allow me to strengthen probability, optimization, adversarial machine learning, security analysis, and research communication while developing the project into a rigorous thesis.

Why a thesis-based MSc

A thesis-based MSc will provide the structured supervision, mathematical depth, research community, and computing environment needed to move from a strong independent baseline to a defensible security study. I hope to work with a supervisor whose expertise in federated learning, adversarial machine learning, privacy, or distributed-system security can guide the threat model and evaluation design.

Long-term objective

My long-term goal is to contribute to intelligent systems that remain dependable when data are sensitive, distributed, heterogeneous, or exposed to adversarial behaviour. I am especially motivated by research that connects machine-learning performance with realistic security assumptions and responsible deployment constraints.